SEEDLINE · PROBLEM STATEMENT

SeedLine problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-24	Amish Chadha	Kit 1.8 pipeline (kit 1.9.1 upgrade): users-and-needs and stakeholders tables (Tables 1 and 2), the source of SDL-URS-001; requirements now in SDL-URS-001 and SDL-PRS-001
0.4	2026-10-02	Amish Chadha	Partner rule and first candidate to approach (SDL-DEC-001, 2026-10-02)
0.3	2026-09-25	Amish Chadha	Record the first user group and row-spacing kit decisions (SDL-DDR-001); update open questions
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, out of scope, prior work with sources)
0.1	2026-09-24	Amish Chadha	Initial scaffold

SeedLine problem statement

Hand-seeding is slow and uneven, and precision planters are too costly for small plots. The gap is a push seeder that places one seed at a time at a set spacing, costs about the same as the cheapest plate seeder, and can be adapted to any crop by printing a new metering plate instead of buying one.

The problem

Most of the world's farms are small. About 84 % of the world's roughly 570 million farms are under 2 ha (Our World in Data, from Lowder et al.). On these plots and on market gardens, seed is still sown by hand with a hoe, a stick or a jab planter, or dribbled from a cheap plate seeder.

Three problems follow:

1. **Labor.** Hand planting with a hoe has been estimated at about 56 h per hectare for one worker in southern Africa, and a jab planter cuts that to about a third (Baudron et al., cited in a CSBE jab planter study). Planting happens in a short window after the first rains, so slow planting delays part of the crop.
2. **Uneven spacing.** Hand-dropped seed lands in clumps and gaps. Uneven stands cost yield: Purdue trials on maize found a loss of about 2.2 to 2.5 bu/acre (about 140 to 160 kg/ha) for each inch (25 mm) increase in the standard deviation of plant-to-plant spacing (Nielsen, Purdue University). Clumped vegetable seed also has to be thinned by hand.
3. **Cost and fit of existing tools.** Low-cost plate seeders such as the Earthway 1001-B (about $187 with six plates) pick up several seeds at a time, so crops usually need thinning, while singulating roller seeders such as the Jang JP-1 cost about $499 before rollers at about $25 each (University of Minnesota Extension, 2024). Users also report that the Earthway is inaccurate with small seed and has no in-row spacing adjustment (Paperpot Co.). A farmer whose crop or seed size does not match a stock plate has no easy way to get one.

SeedLine is a single-row push seeder whose ground wheel drives a vertical metering plate through a chain. The plate is 3D printed for each crop and seed size, so a farmer, a cooperative or a local maker with a printer can make a plate for any seed. An optional row-marker kit sets the spacing between rows.

First users (decided by Amish, 2026-09-25, SDL-DDR-001): smallholder field crops, meaning maize, beans, sorghum and groundnut on 0.75 m rows. Market-garden vegetables follow as a second plate set, with seed under 2 mm sown pelleted. The row-spacing kit is a marker arm.

Users and context

The user requirements in SDL-URS-001 are drawn from Tables 1 and 2. Both are desk research and proposals until co-design sessions with a partner confirm them (see "User research and co-design" below).

Table 1. Users and needs.

User type	Goals and needs	Typical use scenario	Pain points and challenges
Smallholder farmer	Plant maize, beans, sorghum or groundnut at the right spacing and depth in the planting window, with less labour and less seed	Plots of 0.2 to 2 ha, hand-tilled or ox-ploughed soil, rain-fed, often one worker pushing for several hours after the first rains	Hoe planting takes about 56 h/ha; hand-dropped seed clumps; plate seeders drop several seeds per cell; no plate exists for local varieties; cash for tools is short
Market gardener	Sow vegetables in beds without thinning, at low cost	Beds of 0.75 to 1.2 m, fine tilled soil, 3 to 6 rows per bed, pelleted seed for small seed	Thinning labour; the Earthway is inaccurate with small seed and has no in-row adjustment; singulating seeders cost about USD 500 before rollers
Cooperative or agro-dealer with a printer	Print plates on demand for members' crops and varieties; lend or rent seeders	Rural town, one FDM printer, basic tools; members bring seed samples	Needs a plate file per seed size and a way to check cell size; printer time and filament cost
Local welder or maker	Build and repair the frame and drive from common steel tube and bicycle or roller chain parts	Workshop with a welder or a drill, taps and spanners; bicycle and trolley spares in town	Drawings that assume tools or stock they do not have; parts that need a lathe
Extension worker or researcher	Recommend spacings, demonstrate row planting and check stands	Field demonstrations and trial plots, conservation agriculture programmes	Uneven stands hide the effect of spacing advice; no low-cost tool that sets spacing reliably

Table 2. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Co-design and field-trial partner (first candidate: CIMMYT small-scale mechanization work in Eastern and Southern Africa; SDL-DEC-001)	Validates crops, spacings, soils and cost; runs field trials	Co-design sessions and trial plots before the design is frozen; nothing is agreed yet
Seed companies and agro-dealers	Sell seed, could print plates matched to the varieties they sell, rent seeders	Interviews on plate demand and treated-seed handling
Extension services and NGOs	Promote row planting and conservation agriculture; train farmers	Demonstration and training material; feedback on safety notes
Local workshops and maker spaces	Build, repair and adapt the seeder; print plates as a service	Review of the build plan and drawings for local stock and tools
Regulators and standards bodies	Machinery and pesticide-handling rules where the seeder is made or sold	Register in SDL-REG-001; items marked "to assess" need a qualified reviewer
Design Molecule (Amish Chadha)	Owns the open design, its decisions and its licence	Decisions recorded in SDL-DEC-001

Operating environment

- **Soil:** tilled loam to sandy loam, with clods, crop residue and stones up to about 50 mm; seedbed quality varies widely on hand-tilled plots.
- **Seed:** from maize and beans (8 to 15 mm) down to vegetable seed of 1 to 2 mm; some seed is treated with fungicide or insecticide.
- **Climate:** 10 to 40 °C, strong sun, dust; rain and mud at planting time.
- **Supply chain:** steel tube, bicycle chain and sprockets, bearings and wheelbarrow or trolley wheels are common in regional towns; 3D printers are uncommon in villages but increasingly present in towns, universities and maker spaces.

Constraints

- Garage-buildable prototype, about $400 USD (`project.yaml`).
- A replicated base seeder (without the marker kit) should cost no more than a low-cost plate seeder, about $200 in parts.
- Metering plates printable in PLA, PETG or ASA on a common desktop FDM printer, without supports.
- Frame and drive from common steel sections, bicycle chain parts and bearings that local workshops stock.
- Pushed by one person at walking speed on a tilled seedbed.
- No engine, battery or electronics in the base design.

Out of scope

- Tractor or animal-drawn planters.
- Fertilizer placement (a later option, not part of this concept).
- Transplanting, seed-tape or paper-pot systems.
- Seed treatment, seed supply or agronomic advice beyond the plate spacing tables.
- Vacuum or air-assisted metering.

Prior work

- **Commercial push seeders.** The Earthway plate seeder and the Jang roller seeder set the price and performance range for push seeders (UMN Extension; Paperpot Co.). Both use interchangeable plates or rollers bought from the maker.
- **Open 3D-printed seeders.** The Hour Farm published a hackable, 3D-printed precision seeder with interchangeable printed seed discs on FarmHack in 2019 under CC BY 4.0 (FarmHack). It shows that printed discs work for vegetable seed at bed scale.
- **Printed parts in precision metering.** A Norfolk farmer printed singulator discs and sprockets for a six-row vacuum maize planter, reporting about 99 % singulation and no visible wear after 70 ha (3D Printing Industry, 2019). This is a tractor planter, but it supports the durability of printed metering parts.
- **Jab planters.** Hand jab planters are promoted for conservation agriculture in Africa and cut planting time to about a third of hoe planting (CSBE study; FAO jab planter user manual). They place seed hill by hill, so spacing depends on the operator.

- **Research push planters.** Many university studies build and test manual single-row planters with plate or cell-wheel metering (for example, a manually operated planter for different seeds). Their field capacity and seed placement data have not yet been reviewed against SDL-CAL-001; that review remains open.

Open questions

- Which partner and region for co-design and field trials? Decided by Amish, 2026-10-02 (SDL-DEC-001): a partner already working with smallholder maize and bean farmers on 0.75 m rows on ploughed land, ideally with field trial capacity; first candidate to approach, CIMMYT's small-scale mechanization work in Eastern and Southern Africa. Nothing is agreed yet.
- Who prints plates in practice: the farmer, a cooperative, an agro-dealer or a central maker who mails them?
- How common is treated seed among target users, and what handling rules apply?
- Do market gardeners want two or three metering units ganged on one frame (the later bed-seeding variant)?

User research and co-design

Requirements must come from the people who will use the seeder.

- [ ] Identify a local partner organization (farmer cooperative, extension service, NGO or university)
- [ ] Run co-design sessions with intended users; record who, where and what was learned
- [ ] Validate crop list, spacing, soil and cost assumptions in the field
- [ ] Revise the user and product requirements (SDL-URS-001, SDL-PRS-001) from findings before freezing the design